

Quiz 4 [25 min]

If you find XXX, write the correct replacement for XXX.

For the True/False questions, Check **ONLY T** if the given statement in the question is correct or only **F** if not.

Correct Answer 4pts, No Answer -1 pts, Wrong Answer -2 pts. If the total points < 0, your score will be set to 0.

1. $-17 \bmod 2 = \text{XXX}$. / **XXX = 1**

2. $199 \bmod 19 = \text{XXX}$. / **XXX = 9**

3. Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, there are integers s and t with $b = a + sm$ and $d = c + tm$. Hence, $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$. Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. **(T)**

4. Let $n = 2k + 1$ for some integer k . Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$. Since either k or $k + 1$ is even, $4k(k + 1)$ is multiple of 8. Therefore $n^2 - 1$ is multiple of 8, so $n^2 \equiv 1 \pmod{8}$. **(T)**

5. Because $\min(x, y) + \max(x, y) = x + y$, the exponent of p_i in the prime factorization of $\gcd(a, b) \cdot \text{lcm}(a, b)$ is the sum of the exponents of p_i in the prime factorizations of a and b . Therefore, if a and b are positive integers, then $ab \equiv \gcd(a, b) \cdot \text{lcm}(a, b)$. **(T)**

6. There are infinitely many primes of the form $4k + 3$, where k is a nonnegative integer. Suppose that there are only finitely many primes of the form $4k + 3$, namely, $q_1, q_2, q_3, \dots, q_n$, where $q_1 = 3$, $q_2 = 7$, and so on. Let $Q = 4q_1q_2 \dots q_n - 1$. If Q is prime, then we have found a prime of the desired form different from all those listed. If Q is not prime, then Q has at least one prime factor not in the list $q_1, q_2, \dots, q_n$. Because all odd primes are either of the form $4k + 1$ or of the form $4k + 3$, and the product of primes of the form $4k + 1$ is also of this form (because $(4k + 1)(4m + 1) = 4(4km + k + m) + 1$), there must be a factor of Q of the form $4k + 3$ different from the primes we listed. **(T)**

7. An XXX is a finite set of precise instructions for performing a computation or for solving a problem.

XXX = algorithm

8. The constants C and k are called XXX to the relationship $f(x)$ is $O(g(x))$.

XXX = witnesses

9. The least integer n such that $f(x)$ is $O(x^n)$ for the function $f(x) = 3x^5 + (\log x)^4$ is XXX. / **XXX = 5**

10. The least integer n such that $f(x)$ is $O(x^n)$ for the function $f(x) = (x^4 + x^2 + 1)(x^4 + 1)$ is XXX. / **XXX = 0**

11. If we use the bubble sort to sort 3, 1, 5, 7, 4, the lists obtained at the third step or pass is XXX.

XXX = 1, 3, 4, 5, 7;

At the end of the first pass: 1, 3, 5, 4, 7;

At the end of the second pass: 1, 3, 4, 5, 7;

At the end of the third pass: 1, 3, 4, 5, 7;

At the end of the fourth pass: 1, 3, 4, 5, 7;

12. If $f_1(x)$ and $f_2(x)$ are functions from the set of positive integers to the set of positive real numbers and $f_1(x)$ and $f_2(x)$ are both $\Theta(g(x))$, is $(f_1 - f_2)(x)$ also $\Theta(g(x))$. **(F)**

13. Let $f_1(x) = x^2 + 2x$, $f_2(x) = x^2 + x$, and $g(x) = x^2$. Then $f_1(x)$ and $f_2(x)$ are both $O(g(x))$, but $(f_1 - f_2)(x)$ is not. **(F)**